

LU3: Digital Business Models & Ecosystems

Architecting Platform Multi-Sided Dynamics, SaaS Funnels, API Ecosystems,
and Enterprise Business Transformation



Duration: 3 Hours (180 Mins)



Break: 15 Mins

Overall Unit Learning Outcomes

LU3 ROADMAP

LO1: Multi-Sided Platforms

Analyze multi-sided platform architecture and two-sided market dynamics in businesses such as Uber, Airbnb, and Amazon.

LO2: Recurring Revenue Structures

Evaluate recurring revenue structures including SaaS subscription tiers, freemium conversion funnels, and retention mechanics.

LO3: API-Driven Partner Networks

Map API-driven partner networks and platform ecosystems that enable cross-industry co-creation of value.

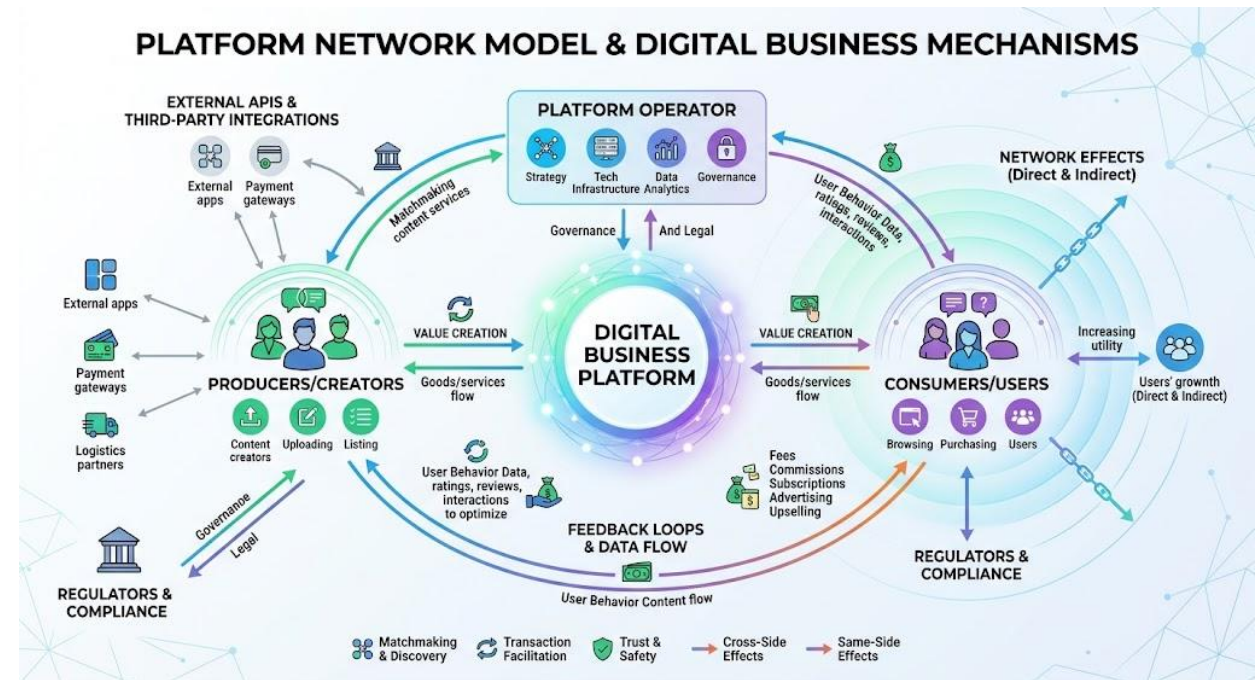
LO4: Business Model Transformation

Compare strategic paths between digital-native startups and legacy enterprises undergoing business model transformation.

LO1: Multi-Sided Platform Architecture

THEORY

- ✓ **Two-Sided Market Mechanics:** Connecting distinct user groups (buyers & sellers, drivers & riders) to facilitate direct value exchange.
- ✓ **Network Effects:** Harnessing positive feedback loops where increased participation on one side directly drives value for the other.
- ✓ **Platform Orchestration:** Lowering transaction friction via match-making algorithms, automated pricing, and digital trust mechanics.
- ✓ **Market Champions:** Industry benchmarks including Uber (mobility), Airbnb (hospitality), and Amazon Marketplace (e-commerce).



LO1 Activity: Platform Dynamics Workshop

WORKSHOP • 35 MINS



1. Platform Mapping

Select a legacy business and map out a proposed two-sided marketplace architecture identifying primary producers and consumers.

Group Breakout

10 Mins



2. Network Effect Audit

Identify positive and negative direct/indirect network loops. Formulate rules to solve the chicken-and-egg cold start problem.

Analysis & Design

15 Mins



3. Cohort Debrief

Present platform monetization mechanics (take-rates, listing fees, premium matching) to the class for peer evaluation.

Presentation

10 Mins

LO2: Recurring Revenue Structures

THEORY

- ✓ **SaaS Subscription Tiers:** Structuring feature-gated plans (Basic, Pro, Enterprise) aligned with value metrics and customer willingness to pay.
- ✓ **Freemium Conversion Funnels:** Designing frictionless onboarding triggers that convert free tier users into paying subscribers.
- ✓ **Retention & Unit Economics:** Optimizing Customer Acquisition Cost (CAC), Lifetime Value (LTV), and minimizing net revenue churn.
- ✓ **Expansion Revenue:** Leveraging usage-based add-ons and seat upgrades to achieve negative net churn growth.



LO2 Activity: SaaS Funnel & Pricing Lab

LAB EXERCISE • 35 MINS



1. Tier Architecture

Design a 3-tier subscription pricing matrix for a B2B SaaS startup, defining value metrics and feature gates.

Matrix Drafting

12 Mins



2. Funnel Audit

Analyze a simulated low-conversion freemium funnel and propose targeted product-led growth (PLG) interventions.

Case Analysis

13 Mins



3. Retention Pitch

Calculate LTV/CAC ratios and pitch a customer retention playbook aimed at reducing annual user churn by 5%.

Pitch & Feedback

10 Mins

15-Minute Intermission

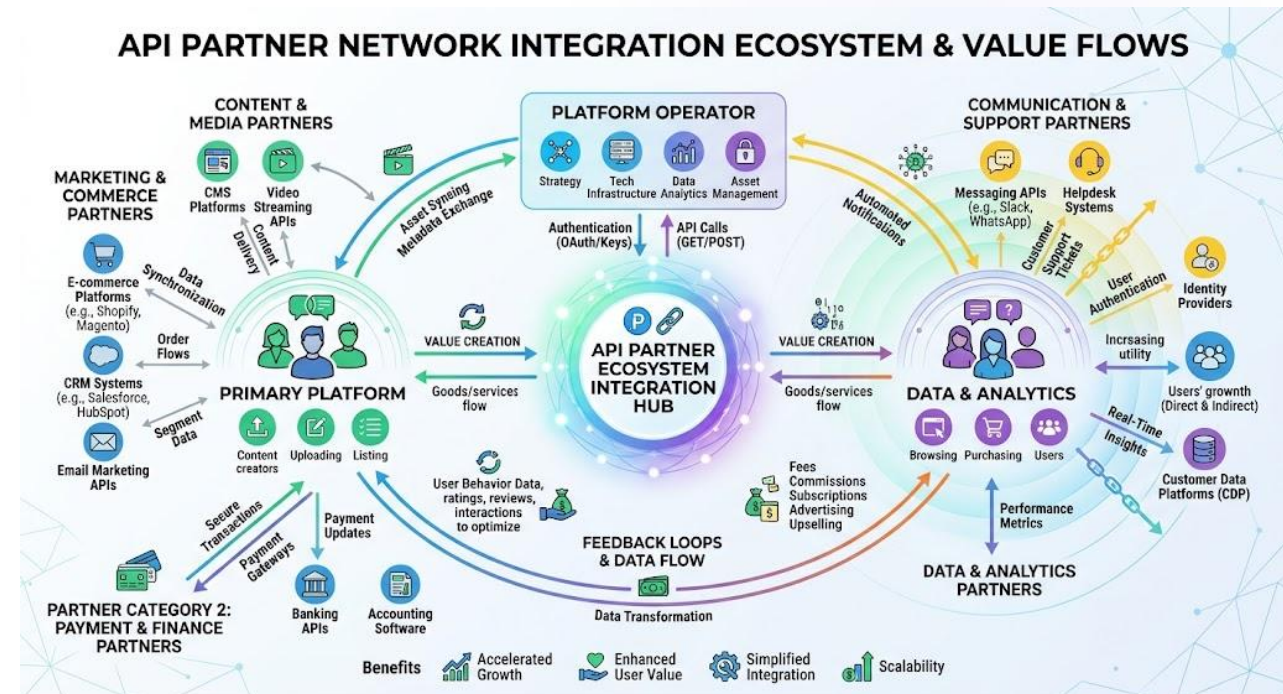
Take a break, grab a refreshment, and reflect on the platform and recurring revenue strategies discussed so far.

Up Next: API-Driven Partner Ecosystems & Legacy Business Model Transformation

LO3: API Networks & Ecosystems

THEORY

- ✓ **API as a Business Model:** Exposing core capabilities (payments, logistics, data) via APIs to create modular integration hubs.
- ✓ **Partner Ecosystem Networks:** Enabling third-party developers to build complementary services on top of core platform infrastructure.
- ✓ **Cross-Industry Co-Creation:** Facilitating collaborative value creation across banking, retail, healthcare, and telecom sectors.
- ✓ **Ecosystem Governance:** Balancing open access with security, data compliance, and revenue-sharing models.



LO3 Activity: API Ecosystem Mapping

WORKSHOP • 30 MINS



1. API Identification

Map out core internal capabilities that can be monetized externally via open or partner-restricted APIs.

Capability Mapping

10 Mins



2. Co-Creation Canvas

Design a cross-industry integration scenario (e.g., Fintech + Healthtech) detailing shared value streams.

Ecosystem Canvas

12 Mins



3. Ecosystem Pitch

Present API partner governance principles and monetization strategies to the peer cohort.

Cohort Review

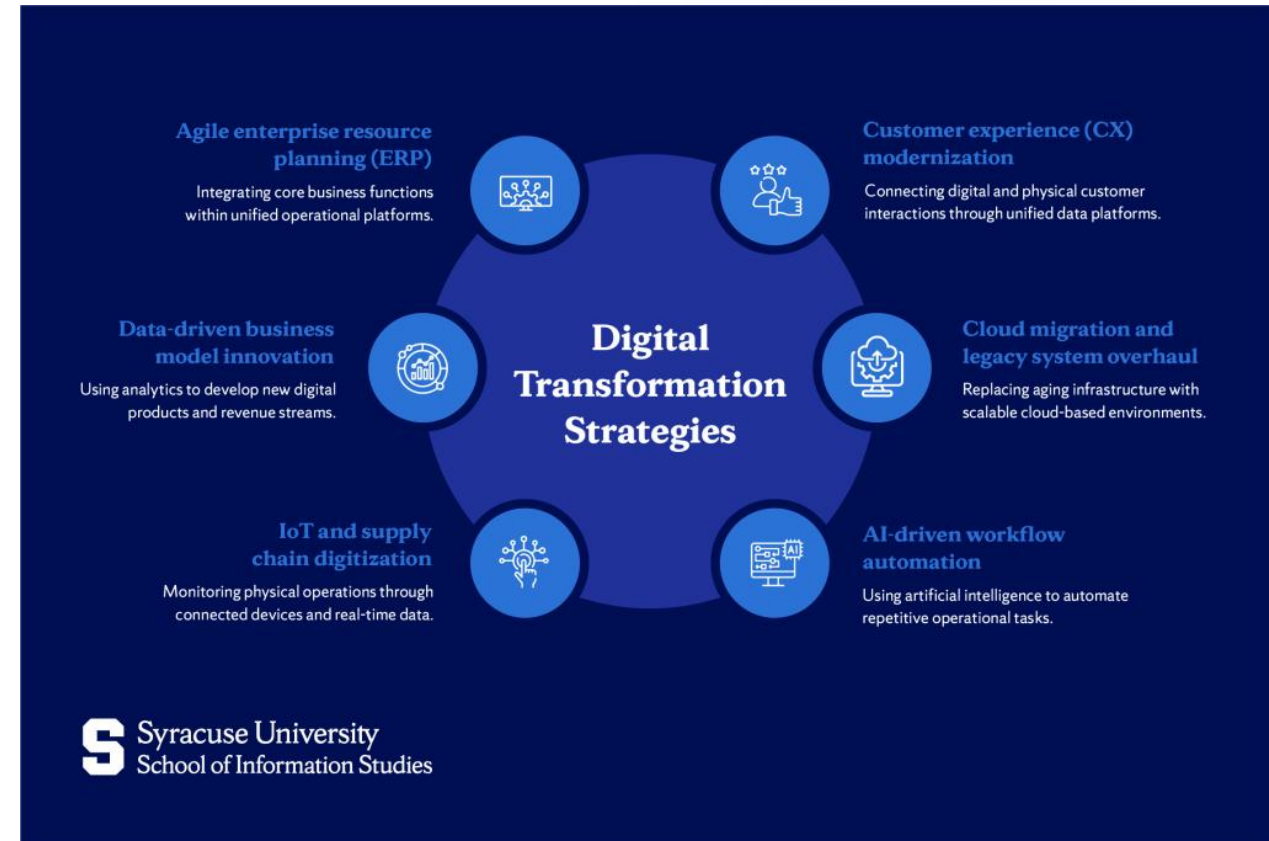
8 Mins

LO4: Business Transformation

Digital-Native vs. Legacy Enterprise

Digital-native startups operate with zero legacy tech debt, agile organizational structures, and product-led growth strategies built for cloud-scale.

In contrast, legacy enterprises undergoing business model transformation face legacy IT infrastructure, cultural inertia, and channel conflict. Success requires ambidextrous innovation—optimizing the core engine while scaling new digital models.



LO4 Activity: Transformation Challenge

SIMULATION • 35 MINS



1. Legacy Debt Audit

Assess a traditional enterprise's business model barriers (channel conflict, legacy systems, margin pressure).

Diagnostic Stage

10 Mins



2. Dual-Track Strategy

Formulate a strategic roadmap balancing core business optimization with new platform/subscription revenue streams.

Strategy Drafting

15 Mins



3. Executive Defense

Defend your business transformation blueprint against board-level objections in a simulated executive review.

Roleplay & Debate

10 Mins

Questions & Summary

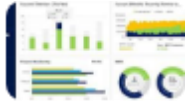
Thank you for participating in LU3: Digital Business Models and Ecosystems.

Image Sources



https://www.slideteam.net/media/catalog/product/cache/1280x720/c/o/conceptual_framework_for_business_digital_transformation_slide01.jpg

Source: www.slideteam.net



https://www.slideteam.net/media/catalog/product/cache/1280x720/s/a/saas_metrics_dashboard_illustrating_revenue_growth_rate_slide01.jpg

Source: www.slideteam.net



<https://techafricanews.com/wp-content/uploads/2025/11/API.png>

Source: techafricanews.com



<https://ischool.syracuse.edu/wp-content/uploads/digital-transformation-business-strategy-1024x683.png>

Source: ischool.syracuse.edu